A Report On The Calculation Of The Excited States Of Z_azobenzene At The DFT/def2-SVP Level

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Abstract

The calculation of excited states for the system 'Z_azobenzene' is presented, accompanied by automated analysis and image generation provided by the Digichem software package. The calculation was performed using the ORCA software package at the DFT/def2-SVP level of theory. The total self-consistent field (SCF) energy of the system was found to be -15565.36 eV after 1 step. The highest-occupied molecular orbital (HOMO) and lowest-unoccupied molecular orbital (LUMO) were calculated to be -5.47 and -2.33 eV respectively, corresponding to a HOMO-LUMO band gap of 3.14 eV. The permanent dipole moment (PDM) was calculated to be 3.37 D. In total, 2 excited states were calculated with singlet multiplicity. The most intense absorption peaks were calculated to be at 323 and 626 nm. The lowest energy singlet excited state (S_1) was calculated to be 1.95 eV (635 nm).

Table 1: *Summary of overall calculation metadata.*

Date[a] (Duration[b])	CPUs (Memory)	Success (Converged)	Computational package	Level of theory	Solvent (model)	Calculations	Wavefunction	Multiplicity	T[c] / K	P[d] / atm
N/A (3 m, 3 s)	1 (4 GB)	True (N/A)	ORCA (5.0.3)	DFT/def2-SVP	Gas-phase (N/A)	Excited States	restricted	1 (singlet)	N/A	N/A

[a]: The date and time at which the calculation was completed. [b]: Total combined duration in real-time (wall-time) for all components of the calculation. [c]: Temperature used for thermochemistry analysis. [d]: Pressure used for thermochemistry analysis.

Summary Of Results

SCF Energy

Table 2: *Summary of SCF energy properties.*

No. of steps	1
Final energy	-15565.3621 eV
Final energy	-1,501,829 kJ·mol^{-1}

Geometry

Table 3: *Summary of geometry properties.*

Formula	$C_{12}H_{10}N_2$
SMILES	c1ccc(/N=N\c2ccccc2)cc1
Exact mass	182.2260 g·mol^{-1}
Molar mass	182.2212 g·mol^{-1}
Alignment method	Minimal
X extension	8.56 Å
Y extension	4.95 Å
Z extension	4.38 Å
Linearity ratio	0.42
Planarity ratio	0.12

Molecular Orbitals

Table 4: *Summary of HOMO & LUMO properties.*

$E_{HOMO,LUMO}$	3.14 eV
E_{HOMO}	-5.47 eV
E_{LUMO}	-2.33 eV

Permanent Dipole Moment

Table 5: *Summary of the permanent dipole moment properties.*

Total	3.37 D
X axis angle	24.93 °
XY plane angle	0.58 °

Excited States

Table 6: *Summary of the calculated excited states. E_x: The energy of excited state x. λ_x: The wavelength of a photon of equivalent energy to excited state x. f_x: The oscillator strength of the excited state transition x. ΔE_{xy}: The difference in energy between the lowest excited states of multiplicity x and y.*

No. calculated singlets	2
E_{S_1}	1.95 eV
λ_{S_1} **(colour, CIE)**	635 nm (Red, (0.71, 0.29))
f_{S_1}	0.03
Simulated Absorption Peaks	323 and 626 nm

Methodology

Metadata

The calculation of the excited states was performed using the **ORCA (5.0.3)** program, the **DFT** method and the **def2-SVP** basis set. It was completed after a total duration of **3 m, 3 s** and **finished successfully**. The base multiplicity of the system under study was **1 (singlet)**. Finally, a **restricted wavefunction** was used, resulting in a single set of doubly occupied orbitals. The full calculation metadata is tabulated in table 1.

Analysis

The report presented here was generated using the Digichem software package. This toolset relies upon a number of third-party applications and libraries which should be cited appropriately in derivative works. In particular, the calculation results described within were parsed by the cclib library.[1] Scientific constants which were used, among other things, for the interconversion of scientific units were provided by SciPy.[2] Commission internationale de l'éclairage (CIE) coordinates, along with visual representations of the equivalent colour, were calculated using the Colour Science library.[3] Three-dimensional plots of atom positions and calculated densities, including

molecular orbitals, were rendered using Visual Molecular Dynamics (VMD)[4] and the Tachyon ray-tracer.[5] Finally, two-dimensional graphs were plotted using the MatPlotlib library,[6] while this report itself was prepared using the Mako template library[7] and the Weasyprint library[8], the latter of which was responsible for generation of the PDF file.

Discussion

Total SCF Energy

The total energy of the system was calculated at the **self-consistent field (SCF)** level, corresponding to the energy calculated by the density-functional theory (DFT) method, with a value of -15565.36 eV, corresponding to -1,501,829 $KJmol^{-1}$. A plot of the total SCF electron density is shown in figure 1.

A: *B:*
C: *D:*

Figure 1: *Plot of the total SCF electron density, plotted with an isovalue of 0.02. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

Geometry

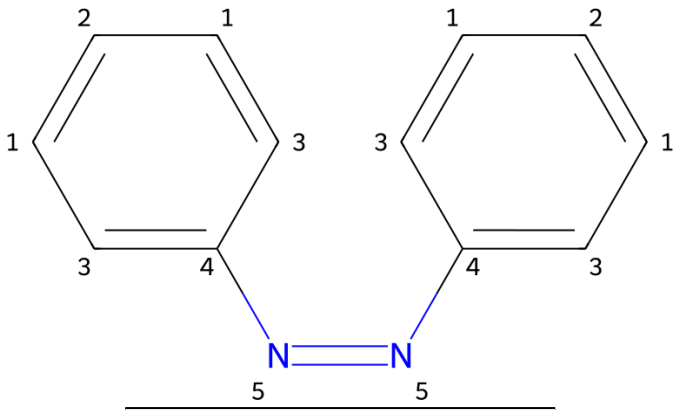

Figure 2: *Labelled structure of Z_azobenzene.*

The **empirical formula** of the studied system was $C_{12}H_{10}N_2$, corresponding to a **molecular mass** of 182.22 $gmol^{-1}$ and an **exact mass**, considering only specific atomic isotopes, of 182.23 $gmol^{-1}$. The molecular structure, with atom labelling, is shown in figure 2. The molecular geometry was aligned to the cartesian (X, Y and Z) axes by the **Minimal (MIN)** method, and the resulting atomic position are displayed in figure 3. Using this method, the **extent of the molecular system** in the X, Y and Z axes (L_X, L_Y and L_Z, corresponding to the molecular width, length and height respectively) was determined to be 8.56, 4.95 and 4.38 Å respectively. These extensions give rise to a **molecular linearity ratio** ($1-(L_Y/L_X)$) and **planarity ratio** ($1-(L_X/L_Y)$) of 0.42 and 0.12 respectively.

A: *B:*
C: *D:*

Figure 3: *The molecular structure, aligned using the Minimal (MIN) method. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

Permanent Dipole Moment

The calculated **permanent dipole moment** was 3.37 D, with a vector (x,y,z) of -1.34, -2.50, -1.82 D. The origin point of the dipole vector was (-12.70, 2.78, -1.69). The angle between the dipole moment vector and the x-axis was 24.93 °, while the angle between the dipole moment and the xy-plane was 0.58 °. A plot of the permanent dipole moment is shown in figure 4.

A: *B:*
C: *D:*

Figure 4: *The permanent dipole moment (red arrow) plotted against the aligned molecular geometry with a scale of 1 Å = 1.0 D. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

Molecular Orbitals

In total, 246 doubly occupied molecular orbitals were calculated, divided into 48 occupied orbitals and 198 unoccupied (or virtual) orbitals. The calculated energies of the **HOMO and LUMO** were -5.47 and -2.33 eV respectively, corresponding to a **HOMO-LUMO band gap** of 3.14 eV (figure 8). Plots of the orbital density for the HOMO and LUMO are shown in figures 5-6 respectively, while the orbital overlap between the HOMO and LUMO is shown in figure 7.

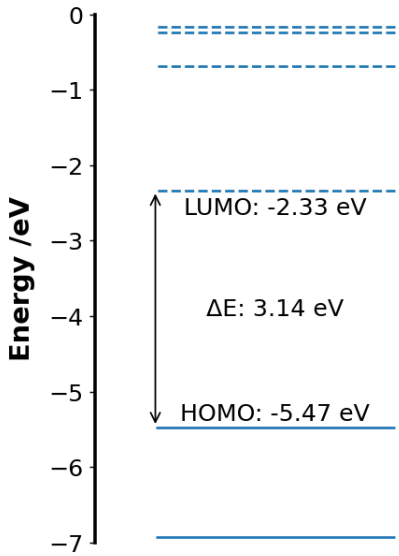

Figure 8: *Graph of the calculated molecular orbital energies in close proximity to the HOMO-LUMO gap. Solid lines: occupied orbitals, dashed lines: virtual orbitals.*

A: *B:*
C: *D:*

Figure 5: *Orbital density plots of the HOMO, plotted with isovalue: 0.02. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

A: *B:*
C: *D:*

Figure 6: *Orbital density plots of the LUMO, plotted with isovalue: 0.02. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

A: *B:*
C: *D:*

Figure 7: *Orbital density plots of the HOMO (blue) and LUMO (red), plotted simultaneously with isovalue: 0.02. A: In the X/Y plane, B: In the X/Z plane, C: In the Z/Y plane, D: 45° to the axes.*

Excited States

In total, the energies of two singlet electronic excited states were calculated, which are shown in figure 9. The energy of the

lowest **singlet excited state (S_1)** was 1.95 eV, corresponding to absorption by a photon with a wavelength of 635 nm, a red 'color' ■ and CIE coordinates of (0.71, 0.29). A complete table of the calculated excited state properties is available in table 7. In addition, an electronic transition spectrum was simulated using a gaussian function with full-width at half maximum (FWHM) of 0.40 eV, from which the **two most intense peaks** were found at 323 and 626 nm. The full simulated absorption spectrum is shown in figure 10.

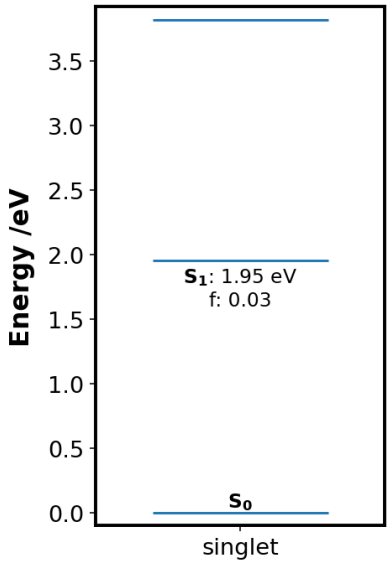

Figure 9: *Graph of the calculated excited states. f: oscillator strength of the relevant ground to excited state transition.*

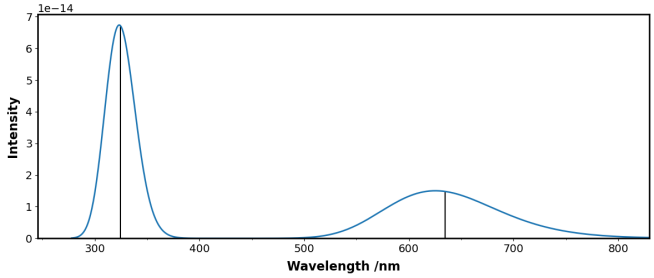

Figure 10: *Graph of the simulated absorption spectrum. Excited states are shown as vertical black lines, while peaks simulated with a gaussian function with FWHM: 0.40 eV are shown as a blue line. Peaks can be found at: 323 and 626 nm.*

Tables Of Results

Atom Coordinates

Table 8: *Coordinates of the atoms of the system under study, as aligned to the cartesian axes by the Minimal method. Atoms that are chemically equivalent have been assigned the same group number.s*

Index	Element	Group	X Coord /Å	Y Coord /Å	Z Coord /Å
1	C	C_1	-10.1708900	2.3468000	-0.1063700
2	C	C_2	-10.1954900	0.9524000	-0.1859600
3	C	C_1	-9.0051400	0.2271600	-0.0946900
4	C	C_3	-7.7874500	0.8948800	0.0685700
5	C	C_4	-7.7515100	2.3033200	0.1307500
6	C	C_3	-8.9552100	3.0180600	0.0574100
7	H	H_3	-8.9488700	4.0990700	0.1214500
8	H	H_1	-11.0942200	2.9080400	-0.1692500
9	H	H_2	-11.1372200	0.4337500	-0.3108400
10	H	H_1	-9.0261700	-0.8538200	-0.1453300
11	H	H_3	-6.8828300	0.3086000	0.1542400
12	N	N_5	-6.5392700	3.0321100	0.3211700
13	N	N_5	-5.3935900	2.7043800	-0.0758500
14	C	C_4	-5.0806400	1.7618700	-1.1008600
15	C	C_3	-3.8688200	1.0622200	-1.0165500
16	C	C_1	-3.5080100	0.1447300	-2.0081800
17	C	C_2	-4.3507200	-0.0727300	-3.1008100
18	C	C_1	-5.5519300	0.6323100	-3.2067700
19	C	C_3	-5.9164400	1.5502100	-2.2168900
20	H	H_3	-6.8430900	2.0955500	-2.3323300
21	H	H_3	-3.2058300	1.2251600	-0.1759500
22	H	H_1	-2.5736200	-0.3959500	-1.9303600
23	H	H_2	-4.0695600	-0.7817400	-3.8687300
24	H	H_1	-6.1991300	0.4714100	-4.0592800

Molecular Orbitals

Table 9: *Energies of the calculated molecular orbitals.*

Level	Label	Symmetry	Energy /eV
64	LUMO+15	A	4.1643
63	LUMO+14	A	4.0485
62	LUMO+13	A	3.9151
61	LUMO+12	A	3.8497
60	LUMO+11	A	3.7750
59	LUMO+10	A	3.1681
58	LUMO+9	A	2.9915
57	LUMO+8	A	2.6965
56	LUMO+7	A	2.6415
55	LUMO+6	A	1.7368
54	LUMO+5	A	1.6647
53	LUMO+4	A	0.6794
52	LUMO+3	A	-0.1678
51	LUMO+2	A	-0.2440
50	LUMO+1	A	-0.6834
49	**LUMO**	**A**	**-2.3341**
48	**HOMO**	**A**	**-5.4749**
47	HOMO-1	A	-6.9267
46	HOMO-2	A	-7.1898
45	HOMO-3	A	-7.2420
44	HOMO-4	A	-7.4279
43	HOMO-5	A	-9.3247
42	HOMO-6	A	-9.4962
41	HOMO-7	A	-9.9787
40	HOMO-8	A	-10.0291
39	HOMO-9	A	-10.1349
38	HOMO-10	A	-10.1962
37	HOMO-11	A	-10.8822
36	HOMO-12	A	-10.9820
35	HOMO-13	A	-11.7472
34	HOMO-14	A	-11.7800
33	HOMO-15	A	-12.1605

Excited States

Table 7: *Energies and other properties of the calculated excited states.*

Number	Symbol	Symmetry	Energy /eV	Wavelength /nm	Colour (CIE x,y)	Oscillator Strength	Transitions (Probability)
1	S_1	Singlet	1.9536	634.65	Red (0.71, 0.29)	0.0300	
2	S_2	Singlet	3.8188	324.67	Ultraviolet (0.00, 0.00)	0.0356	

References

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